Data Science: SQL Assignment

SQL and DBeaver

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```
# Load required libraries
#| warning: FALSE
#| output: FALSE
invisible(lapply(c("DBI", "duckdb", "nycflights13"),
                 library, character.only=TRUE))
```

```
# Establish database connection
#| warning: FALSE
#| output: FALSE
connection <- DBI::dbConnect(drv = duckdb::duckdb(),
                             dbdir = "./data/flights.duckdb")
```

```
duckdb is storing downloaded extensions and secrets under ~/.duckdb:
i /Users/zakz/.duckdb
This persists across sessions and is shared with the DuckDB CLI and other clients.
i Run duckdb(shared_home = FALSE) to use a temporary directory instead.
i See ?duckdb_storage for details and alternatives.
```

Part I: Answer the following question by running SQL using RStudio.

Section 1.a: List the first 10 rows of the flights table.

We select all the rows with the asterisk symbol and limit our output to the first 10 rows.

```
# Get the first ten rows of the flights table
dbGetQuery(connection, "SELECT * FROM flights LIMIT 10")
```

```
   year month day dep_time sched_dep_time dep_delay arr_time sched_arr_time
1  2013     1   1      517            515         2      830            819
2  2013     1   1      533            529         4      850            830
3  2013     1   1      542            540         2      923            850
4  2013     1   1      544            545        -1     1004           1022
5  2013     1   1      554            600        -6      812            837
6  2013     1   1      554            558        -4      740            728
7  2013     1   1      555            600        -5      913            854
8  2013     1   1      557            600        -3      709            723
9  2013     1   1      557            600        -3      838            846
10 2013     1   1      558            600        -2      753            745
   arr_delay carrier flight tailnum origin dest air_time distance hour minute
1         11      UA   1545  N14228    EWR  IAH      227     1400    5     15
2         20      UA   1714  N24211    LGA  IAH      227     1416    5     29
3         33      AA   1141  N619AA    JFK  MIA      160     1089    5     40
4        -18      B6    725  N804JB    JFK  BQN      183     1576    5     45
5        -25      DL    461  N668DN    LGA  ATL      116      762    6      0
6         12      UA   1696  N39463    EWR  ORD      150      719    5     58
7         19      B6    507  N516JB    EWR  FLL      158     1065    6      0
8        -14      EV   5708  N829AS    LGA  IAD       53      229    6      0
9         -8      B6     79  N593JB    JFK  MCO      140      944    6      0
10         8      AA    301  N3ALAA    LGA  ORD      138      733    6      0
             time_hour
1  2013-01-01 10:00:00
2  2013-01-01 10:00:00
3  2013-01-01 10:00:00
4  2013-01-01 10:00:00
5  2013-01-01 11:00:00
6  2013-01-01 10:00:00
7  2013-01-01 11:00:00
8  2013-01-01 11:00:00
9  2013-01-01 11:00:00
10 2013-01-01 11:00:00
```

Section 1.b: Find the number of flights

i. **in the dataset**

We use the count function to count all the rows in the flights table.

```
# Get the total count of rows in the flights table
dbGetQuery(connection, "SELECT COUNT(*) FROM flights")
```

```
  count_star()
1       336776
```

ii. **for each carrier**

We use the count function to count all the rows in the table, and we select the carrier column from the flights table. Then we group by each unique carrier, and order the table by the grouped counts of rows in descending order.

```
# Get the count of rows for each airline carrier
dbGetQuery(connection, "SELECT COUNT(*), carrier FROM flights GROUP BY carrier ORDER BY COUN
```

```
   count_star() carrier
1         58665      UA
2         54635      B6
3         54173      EV
4         48110      DL
5         32729      AA
6         26397      MQ
7         20536      US
8         18460      9E
9         12275      WN
10         5162      VX
11         3260      FL
12          714      AS
13          685      F9
14          601      YV
15          342      HA
16           32      OO
```

Section 1.c: Find the average departure delay (dep_delay)

We use the average function on the `dep_delay` column of the flights table calculate the average departure delay.

```
# Get the mean of dep_delay from the flights table
dbGetQuery(connection, "SELECT AVG(dep_delay) FROM flights")
```

```
  avg(dep_delay)
1       12.63907
```

Section 1.d: Find the top 5 destinations (dest) With the most flights.

We use the count function on the destination column and select the destinations from the flights table initially. Then we group by unique destinations and order the table by the counts of destinations in descending order. Finally we limit the ouput to the top five rows.

```
# Group by destinations, count the number of observations for each destination,
# order by the counts in descending order, and limit the output to the top five.
dbGetQuery(connection, "SELECT COUNT(dest), dest FROM flights GROUP BY dest
           ORDER BY count(dest) DESC LIMIT 5")
```

```
  count(dest) dest
1       17283  ORD
2       17215  ATL
3       16174  LAX
4       15508  BOS
5       14082  MCO
```

Section 1.e: Create a table showing:

1. **carrier**
2. **average arrival delay (arr_delay)**

Sort the results from largest to smallest delay. You must close the connection when you are done. Use `dbDisconnect()`

We first select as our columns carrier and use the averge function on `arr_delay` from the flights table. Then we group by carrier and order by the average arrival delay in descending order.

```
# Group by carriers and calculate the mean arrival delay. Then order
# by the means in descending order.
dbGetQuery(connection, "SELECT carrier, AVG(arr_delay) FROM flights
           GROUP BY carrier ORDER BY AVG(arr_delay) DESC")
```

```
   carrier avg(arr_delay)
1       F9     21.9207048
2       FL     20.1159055
3       EV     15.7964311
4       YV     15.5569853
```

```
5       OO     11.9310345
6       MQ     10.7747334
7       WN      9.6491199
8       B6      9.4579733
9       9E      7.3796692
10      UA      3.5580111
11      US      2.1295951
12      VX      1.7644644
13      DL      1.6443409
14      AA      0.3642909
15      HA     -6.9152047
16      AS     -9.9308886
```

```
# Close the database connection
DBI::dbDisconnect(connection)
```

Part II

Section 2.a: List the first 10 rows of the flights table.

We select all the rows from the flights table with the asterisk and then limit our results to the top ten rows.

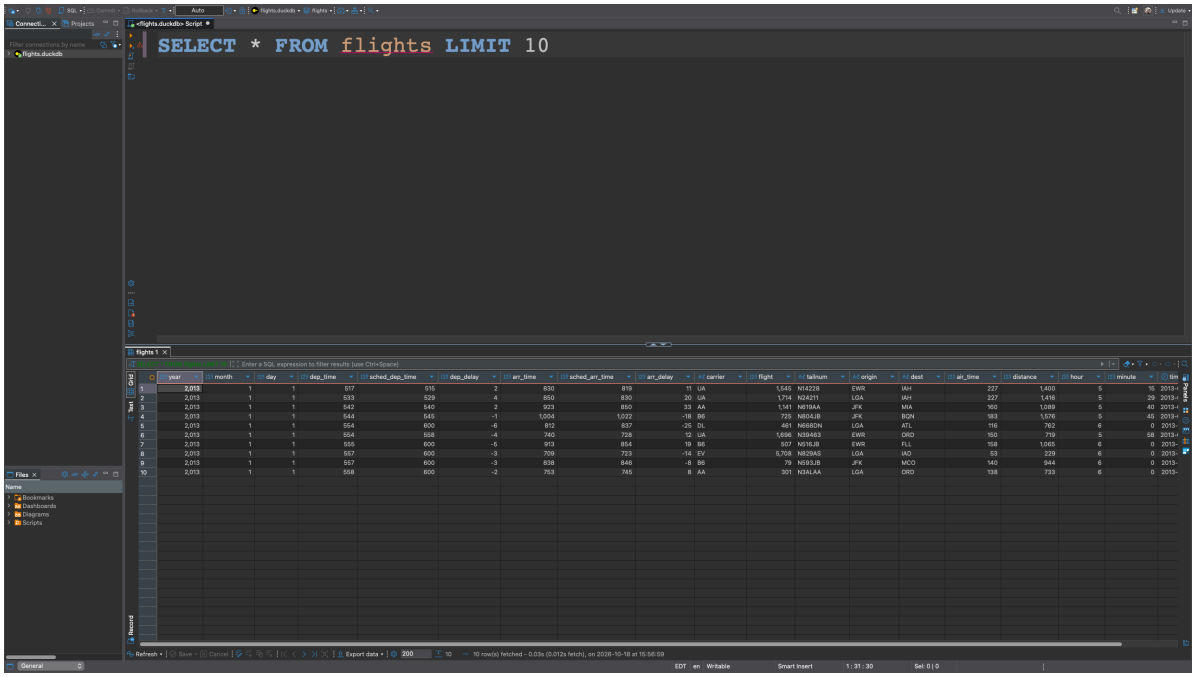


Figure 1: List the first 10 rows of the flights table.

Section 2.b: Count the Number of Flights for Each Carrier

Your output should include:

1. ** carrier**
2. ** number of flights Sort from the largest to smallest.**

We select the counts of all the rows and the carrier column from the flights table. Then we group by carrier, and we order by the row counts in descending order.

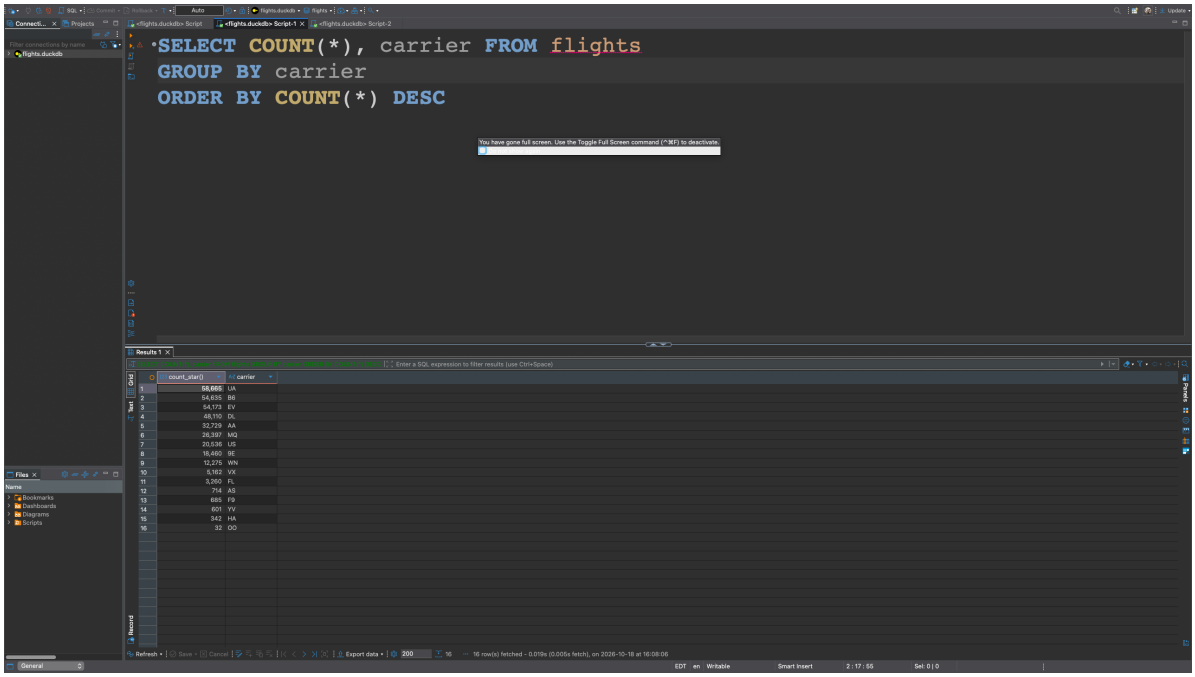


Figure 2: Count the Number of Flights for Each Carrier

Section 2.c: Find the average departure delay by month

Your output should include:

1. **month**
2. **average departure delay**

We first select the average function with the `dep_delay` column and select the month column from the flights table. Then we group by month.

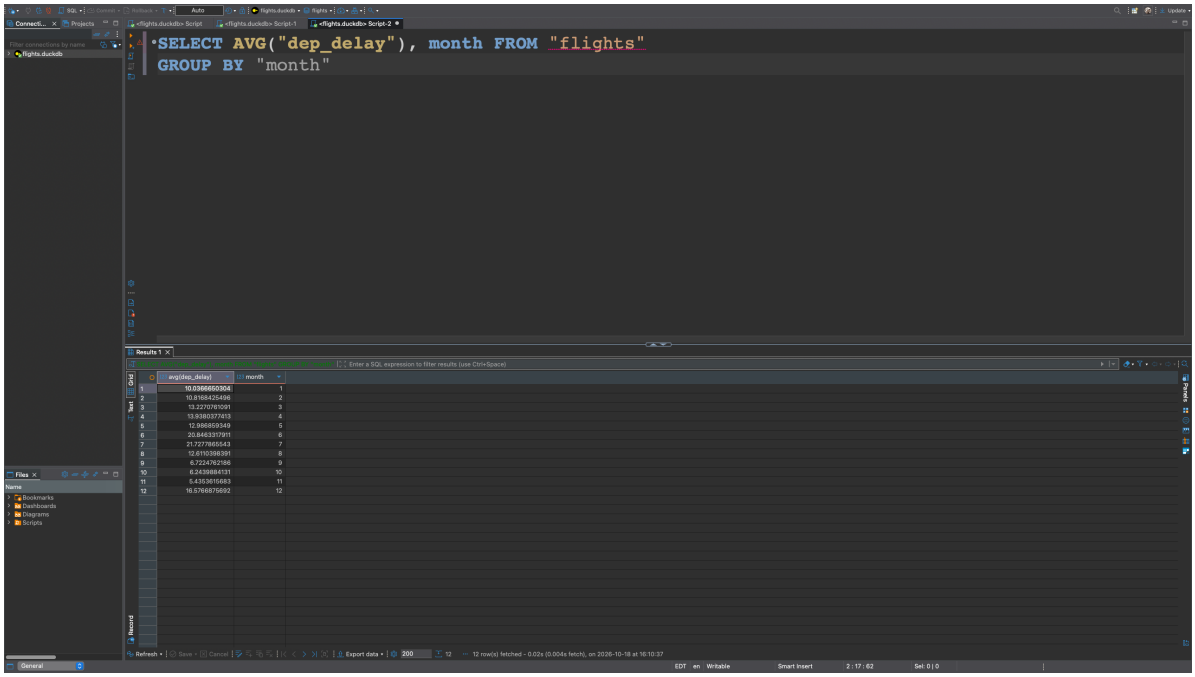


Figure 3: Find the average departure delay by month

Section 2.d: Find the top 10 routes (dest) by:

1. **origin**
2. **dest**

ordered by number of flights.

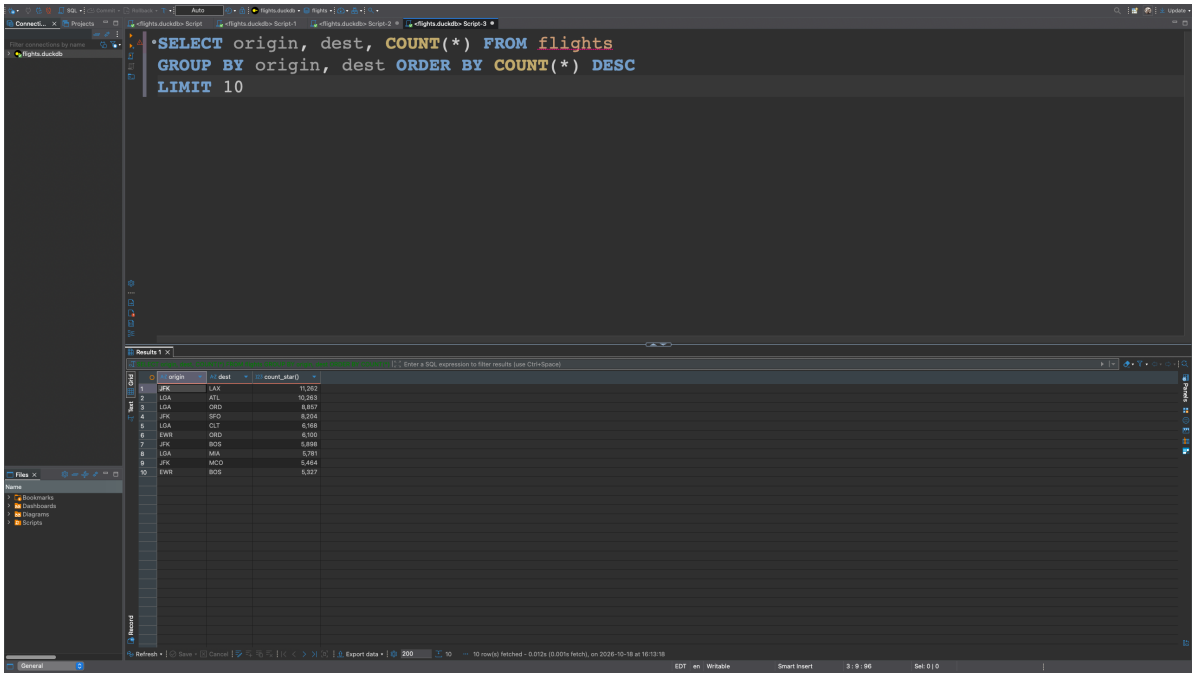


Figure 4: Find the top 10 routes by origin and destination

Section 2.e : Export Part II Section b results

Export the result of part b query to a CSV file using the Export Data feature in DBeaver.

Submit the exported CSV file along with your homework.